

IPSec (Internet Protocol Security)

What is IPSec?

IPSec is a set of protocols used to **secure communication over IP networks**.

It protects data while it travels between two computers or two networks.

IPSec provides:

- **Encryption** Hides data from others
 - **Authentication** Verifies the sender
 - **Integrity** Ensures data is not changed
 - **Anti-Replay Protection** Prevents reuse of old packets
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Why is IPSec Used?

Without IPSec:

PC1 ---- Internet ---- PC2

Data travels in plain text and can be read by attackers.

With IPSec:

PC1 ==== Encrypted Tunnel ==== PC2

Data is encrypted and secure.

Main IPSec Protocols

1. AH (Authentication Header)

Provides:

- Authentication
- Integrity

Does NOT provide:

- Encryption

Anyone can still read the data.

2. ESP (Encapsulating Security Payload)

Provides:

- Encryption
- Authentication
- Integrity

ESP is the most commonly used IPSec protocol.

IPSec Modes

1. Transport Mode

Only the **data (payload)** is encrypted.

IP Header + Encrypted Data

Used for:

- Host-to-Host communication

Advantages:

- Less overhead
 - Faster
-

2. Tunnel Mode

The **entire original IP packet** is encrypted and placed inside a new packet.

New IP Header + Encrypted Original Packet

Used for:

- Site-to-Site VPNs
- Remote Access VPNs

Advantages:

- More secure
 - Most commonly used
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Security Association (SA)

A Security Association is an agreement between two devices about:

- Encryption algorithm
- Authentication algorithm
- Keys to be used

Example:

AES-256
SHA-256
Lifetime = 3600 seconds

IKE (Internet Key Exchange)

IKE automatically:

- Authenticates devices
- Exchanges encryption keys
- Creates Security Associations

Without IKE, key management would be difficult.

IKE Phases

Phase 1

Creates a secure channel between devices.

Checks:

Who are you?
Can I trust you?

Result:

ISAKMP SA Created

Phase 2

Creates the actual IPSec tunnel.

Result:

IPSec SA Created

IPSec VPN Types

Site-to-Site VPN

Connects two office networks.

Office A
||
IPSec Tunnel
||
Office B

Users in both offices can communicate securely.

Remote Access VPN

Connects a remote user to a company network.

Laptop
||
IPSec VPN
||
Company Network

Ports and Protocol Numbers

Service	Port/Protocol
IKE	UDP 500
NAT Traversal (NAT-T)	UDP 4500
ESP	Protocol 50
AH	Protocol 51

How IPSec Works

Step 1

A device wants to send data.

Step 2

IKE negotiates security settings and keys.

Step 3

An IPSec tunnel is established.

Step 4

Data is encrypted.

Step 5

The receiving device decrypts the data.

Real Example

Office A Network:

192.168.1.0/24

Office B Network:

192.168.2.0/24

Using IPSec VPN:

192.168.1.10
||
Encrypted Tunnel
||
192.168.2.10

Both offices can communicate securely over the Internet.

IPSec (Internet Protocol Security) is a security protocol suite used to protect IP communications. It provides encryption, authentication, data integrity, and anti-replay protection. IPSec mainly uses **ESP** and **AH** protocols and works in **Transport Mode** and **Tunnel Mode**. It is widely used for creating secure VPN connections between users and networks.